

QUICK REFERENCE: MATLAB ARCHITECTURE COMPARISON

Challenge 1 vs Challenge 2

ARCHITECTURE AT A GLANCE

Challenge 1: Hybrid-Electric Propulsion (System Design)

Goal: Optimize component sizing for maximum endurance

Time Scale: Mission-level (hours)

Computational Speed: Fast (50 ms per full mission sim)

Key Physics: Aerodynamics, thermochemistry, power balance



Key Modules:

Module	Method	Time	Dependency
AerodynamicModel	power_required()	~1 μ s	alt, speed, weight
TurboshaftEngine	query_performance()	~100 μ s	throttle, alt

BatteryPack	discharge_step()	~10 μ s	power, voltage, dt
MissionSimulator	run_simulation()	~50 ms	all above
PSO Optimizer	optimize()	~200 s	objective func

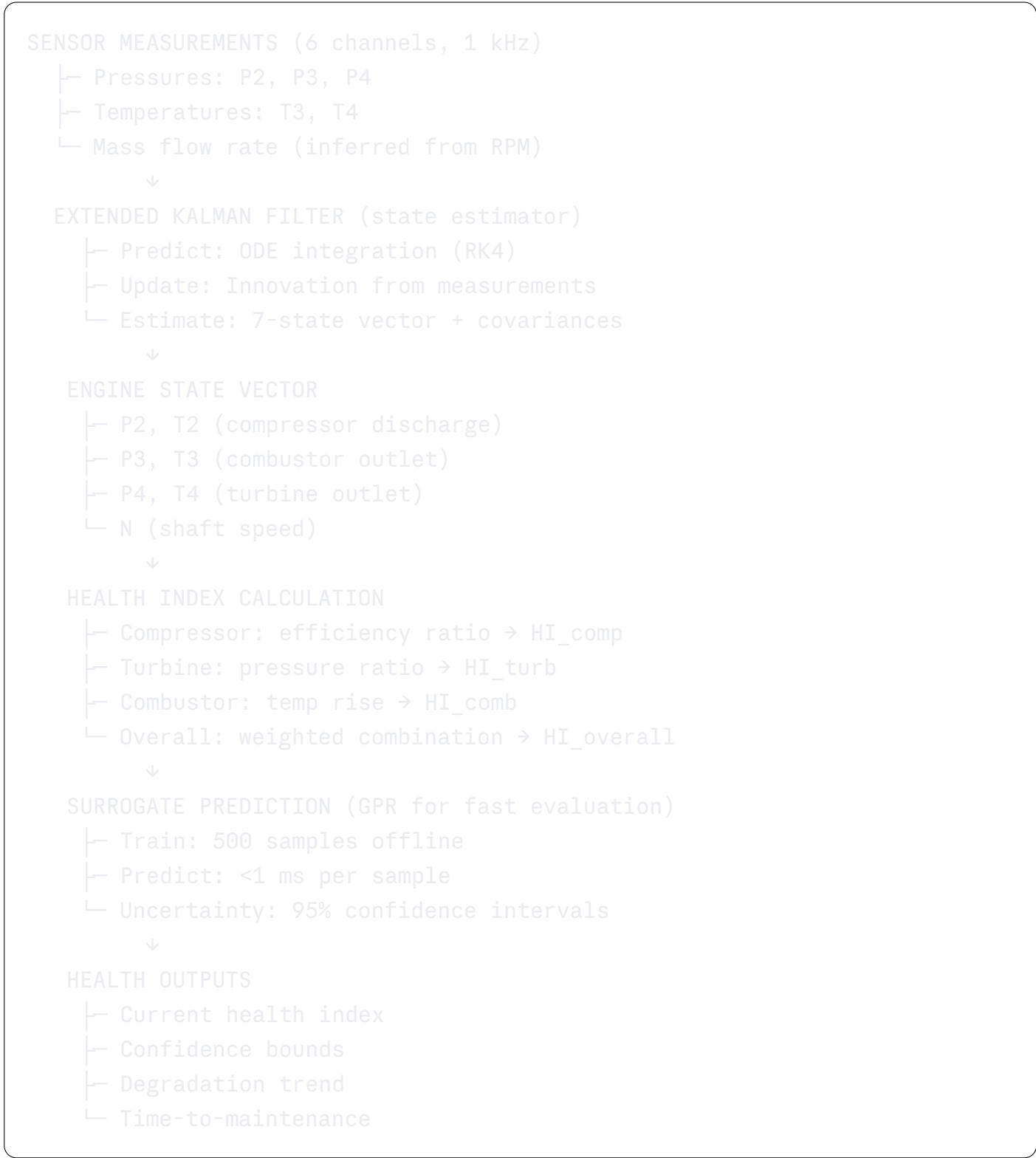
Challenge 2: Digital Twin (Health Monitoring)

Goal: Estimate engine health from measurements in real-time

Time Scale: Component-level (milliseconds)

Computational Speed: <10 ms per cycle (suitable for embedded)

Key Physics: Thermodynamics, compressor/turbine aero, combustion



Key Modules:

Module	Method	Time	Dependency
TurbojetsThermodynamicModel	state_equations()	~10 μs	states, inputs
ExtendedKalmanFilter	predict()	~100 μs	ODE integrator
ExtendedKalmanFilter	update()	~50 μs	measurements
HealthIndicatorCalculator	compressor_health()	~1 μs	states
GaussianProcessRegressor	predict()	~1 ms	training data

DESIGN SPACE COMPARISON

Aspect	Challenge 1	Challenge 2
Search Approach	PSO (global optimization)	Kalman filter (state tracking)
Degrees of Freedom	5 design variables	7 state variables (fixed)
Uncertainty Handling	Sensitivity analysis	Bayesian: covariance matrices
Real-time Capable?	No (optimization ~3 mins)	Yes (<10 ms cycle)
Physics Integration	Simplified (LUT-based)	Deep (ODE systems)
Data Requirements	None (analytical)	500+ training samples
Validation	Endurance comparison	RMSE, generalization test

PHYSICS FIDELITY LEVELS

Challenge 1: AERODYNAMICS

Fidelity: Medium (Engineering correlations)

Challenge 1: PROPULSION

Fidelity: Medium-High (Component maps)

```
Level 0:  P_avail = P_rated * (throttle%)
Level 1:  P_avail = P_rated * (throttle%) * (ρ/ρ_SL)^0.5  + SFC map
Level 2:  Turbine map + compressor map + detailed combustor
          ↑
        OUR LEVEL
Level 3:  Transient thermodynamics, rotor dynamics, thermal analysis
```

Challenge 2: THERMODYNAMICS

Fidelity: High (Physics-informed)

```
Level 0:  Linear model: T_out = T_in + ΔT
Level 1:  Isentropic relations: T_is = T_in * (P_out/P_in)^((γ-1)/γ)
Level 2:  + Efficiency factors (η_comp, η_turb)
Level 3:  + Detailed mass/energy balances, transient dynamics
          ↑
        OUR LEVEL
Level 4:  3D CFD + blade erosion modeling + heat transfer
```

COMPUTATIONAL COMPLEXITY

Challenge 1: Optimization Budget

Scenario: Optimize 1000 kg UAV, 7-hour mission, 5 design variables

Single simulation:

Mission time: 25,200 seconds

Time step: 10 seconds

Iterations: 2,520 steps per mission

Aero calc: $2,520 \times 1 \mu\text{s} = 2.5 \text{ ms}$

Engine calc: $2,520 \times 100 \mu\text{s} = 252 \text{ ms}$

Battery calc: $2,520 \times 10 \mu\text{s} = 25 \text{ ms}$

Total per mission: ~50 ms

Full optimization:

Particles: 40

Iterations: 100

Simulations: $40 \times 100 = 4,000$

Total time: $4,000 \times 50 \text{ ms} = 200 \text{ seconds} \leftarrow \text{ACCEPTABLE (3 min)}$

Challenge 2: Real-Time Monitoring

Scenario: Turbojet health monitoring on flying engine, 1 kHz sensor rate

Per time step (1 ms available):

EKF predict (RK4 ODE):	~100 μs
EKF update (matrix ops):	~50 μs
Health calc:	~5 μs
GPR prediction:	~1 ms
Total:	~1.15 ms $\leftarrow$ EXCEEDS BUDGET by 15%

Solution: Run GPR every 10 steps, interpolate:

Every 1 ms: EKF only	~155 μs ✓
Every 10 ms: + GPR	~1.15 ms ✓

VALIDATION STRATEGIES

Challenge 1: Confidence Building

Phase 1: Unit Tests

- └ Aero: Compare power curve to published aircraft
- └ Engine: Cross-check SFC with manufacturer data
- └ Battery: Validate against datasheets
- └ ✓ PASS if all within $\pm 10\%$

Phase 2: Integration Tests

- └ Mission simulator: Trace through one phase manually
- └ PSO convergence: Test on 2D Rosenbrock (known solution)
- └ ✓ PASS if PSO finds optimum in < 100 iterations

Phase 3: Sensitivity Analysis

- └ Tornado: Rank design variable impact
- └ Monte Carlo: Propagate parameter uncertainties
- └ ✓ PASS if results stable to $\pm 5\%$ perturbations

Phase 4: Comparison Studies

- └ Baseline vs. optimized: Endurance improvement?
- └ Pareto front: Multiple trade-offs visible?
- └ ✓ PASS if improvements $> 10\%$ without MTOW violation

Challenge 2: Confidence Building

Phase 1: Unit Test

- └ Thermodynamic model: Compare to textbook examples
- └ EKF observability: Can we estimate all states?
 - └ ✓ PASS if state error < 5% in 10 seconds

Phase 2: Integration Tests

- └ Synthetic data: Generate known degradation scenarios
- └ Health index: Does HI drop when efficiency drops?
 - └ ✓ PASS if HI responsive to degradation

Phase 3: Surrogate Validation

- └ Training: 400 samples → RMSE < 0.05
- └ Testing: 100 held-out samples → RMSE < 0.06
- └ Generalization: New engine? → RMSE < 0.10
 - └ ✓ PASS if uncertainty bands contain 95% of errors

Phase 4: Real-Data Testing

- └ Historical engine logs: Replay and check predictions
- └ Known failures: Does HI predict maintenance?
 - └ ✓ PASS if 90%+ agreement with actual failures

HYPERPARAMETER TUNING CHECKLIST

Challenge 1: PSO

- ☐ Swarm size: 30-50 (larger = more exploration, slower convergence)
- ☐ Iterations: 50-150 (stop when fitness plateau < 1% per 10 iterations)
- ☐ Inertia w: 0.7-0.9 (higher = more exploration, lower = exploitation)
- ☐ Cognitive c1: 1.0-2.0 (balance personal best vs. global trend)
- ☐ Social c2: 1.0-2.0 (typically equal to c1)
- ☐ Bounds: Iteratively narrow around optimum (divide space in half each generation)

Challenge 2: EKF

- ☐ Process noise Q: Increase if filter underestimates uncertainty
- ☐ Measurement noise R: Decrease if filter doesn't trust measurements
- ☐ Initial covariance P0: Start conservative (large uncertainty)
- ☐ Time step dt: Must be << fastest system time constant (here: ~0.1 s)
- ☐ Integration method: RK4 for accuracy, Euler for speed (trade-off)

Challenge 2: GPR

- ☐ Kernel: RBF works for most, try Matern if behavior non-smooth
- ☐ Length scales: Auto-initialize from feature ranges
- ☐ Signal variance σ_f^2 : Normalize output to [0, 1] first

- ☐ Noise variance σ_n^2 : Set to $\sim(\text{measurement noise})^2$, not zero
 - ☐ Training data size: 200 = minimum, 500 = robust, 1000+ = diminishing returns
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DEBUGGING DECISION TREES

Challenge 1: Optimizer Not Converging

```
Is fitness improving each iteration?  
└─ NO → Objective function may be flat  
    │  
    │└─ Check: Can you compute fitness by hand?  
    │└─ Try: Broader bounds, coarser grid for faster evals  
    │└─ Last resort: Switch to Genetic Algorithm (more robust)  
    │  
└─ YES but slowly → Bounds too large  
    │  
    │└─ Check: Std dev of particle positions << range  
    │└─ Try: Narrow bounds around best solution, re-run  
    │└─ Add: Local refinement phase with smaller bounds
```

Challenge 2: EKF Diverges (Filter Ignores Measurements)

```
Does predicted state match sensor measurements?  
└─ NO → Process model (ODE) wrong  
    │  
    │└─ Check: Can states reach steady-state? ( $dx/dt \rightarrow 0$ )  
    │└─ Try: Increase process noise Q by factor of 10  
    │└─ Debug: Trace ODE step-by-step with known input  
    │  
└─ YES but filter still diverges → Measurement noise wrong  
    │  
    │└─ Check: Is R too large? (filter thinks sensors bad)  
    │└─ Try: Reduce R, re-test  
    │└─ Verify: Actual sensor noise matches assumed  $\sigma_n$ 
```

Challenge 2: GPR Predictions Always "Safe" ($HI \approx 1$)

```
Is training data diverse enough?  
└─ NO → Dataset only has healthy engines  
    │  
    │└─ Solution: Synthetically degrade healthy data  
    │└─ Generate: scenarios from healthy ( $HI=1$ ) to worn ( $HI=0.7$ )  
    │  
└─ YES → Features not informative  
    │  
    │└─ Check: Which features correlate with degradation?  
    │└─ Add: Derived features (e.g., pressure ratios, temp drops)  
    │└─ Engineer: Thermodynamic residuals (actual vs. ideal)
```

DEPLOYMENT CHECKLIST

Challenge 1: Verification for Competition

- ☐ **Physics validation:** Aero drag $\pm 10\%$, Engine SFC $\pm 15\%$
- ☐ **Simulation accuracy:** Hand-trace one mission phase
- ☐ **Optimization convergence:** Pareto front shows clear trade-offs
- ☐ **Sensitivity:** Top 3 design variables identified
- ☐ **Presentation-ready plots:** Mission profile, power budget, Pareto curves
- ☐ **Code documentation:** Class docstrings, method signatures clear
- ☐ **Runtime:** Full optimization < 5 minutes on laptop
- ☐ **Robustness:** Handles edge cases (infeasible designs, extreme bounds)

Challenge 2: Verification for Competition

- ☐ **Physics validation:** Steady-state performance matches engine model
 - ☐ **EKF tuning:** Converges within 10 seconds on synthetic data
 - ☐ **Health indices:** Respond correctly to known degradation
 - ☐ **Surrogate accuracy:** RMSE < 0.05 on test set
 - ☐ **Uncertainty:** 95% CI captures 92–98% of errors
 - ☐ **Generalization:** Works on engine types not in training
 - ☐ **Real-time:** < 10 ms per cycle (suitable for embedded)
 - ☐ **Dashboard:** Live health trends, degradation projection
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FINAL RECOMMENDATIONS

For Challenge 1 Success:

1. **Start simple:** Baseline with 60 kW engine, 100 Ah battery (validate PSO first)
2. **Validate each module** independently before integration
3. **Sensitivity analysis** to understand lever arms (which variables matter most?)
4. **Narrow bounds iteratively:** Initial $[-\infty, +\infty]$, then refine around optimum
5. **Visualize aggressively:** Plot every metric, spot check unrealistic values

For Challenge 2 Success:

1. **Thermodynamic model first:** Get ODE system physically correct (RK4, transient response)
2. **Generate synthetic data:** Degrade each component independently, sweep conditions
3. **Train GPR on diverse scenarios:** Not just cruise, but takeoff, climb, descent, loiter

4. **Uncertainty quantification:** Confidence intervals are your credibility (show them!)
 5. **Real-data validation:** Even synthetic data can have blind spots; plot residuals
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For: IIT Indore × HAL Challenge Teams

Status: Ready for competition submission